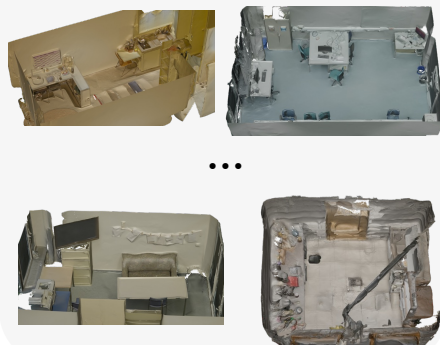
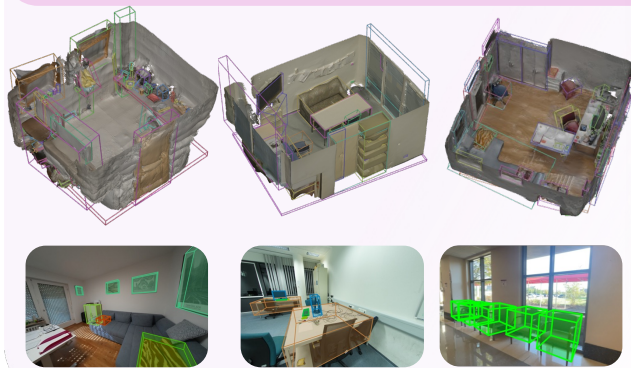


Recon. Mesh



3D Object Detection



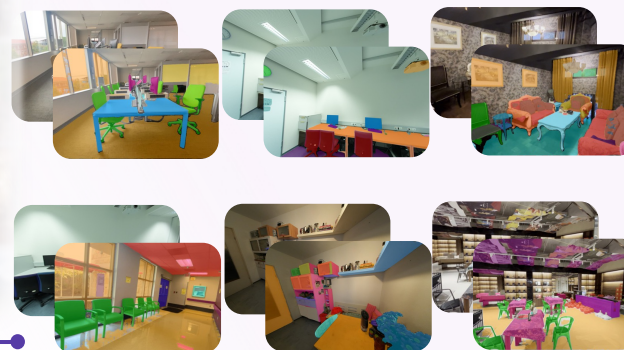
HOLI-SPATIAL

Multi-Modal

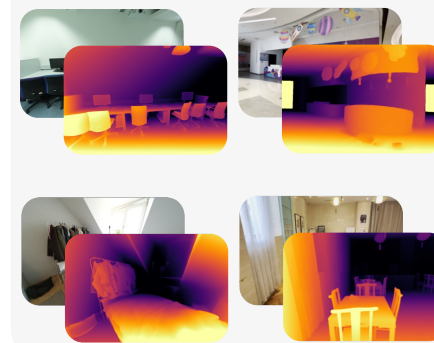
2D-to-3D

4M+ Labels

2D Object Segmentation



Rendered Depth



3D Grounding



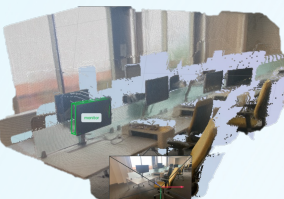
"White vertical panel radiator with horizontal fins, smooth surface"



"Vibrant red fabric sofa with two blue patterned throw pillows"



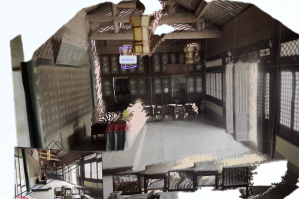
"Vibrant red fabric sofa with two blue patterned throw pillows"



"dark grey monitor with silver frame, black screen, and visible Dell logo on back"



"A poster mounted beside the windows shows a person's face"

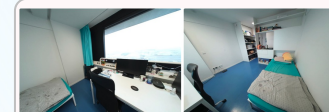


"wooden lantern with square frame and light brown color"

Spatial Reasoning



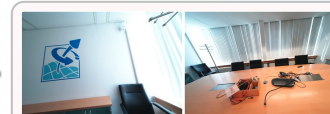
You are positioned at whiteboard and face door. In which direction is radiator relative to you?
A. Front B. **Back-Left**
C. Front-Right D. Front-Left



You are positioned at desk and face suitcase. In which direction is book relative to you?
A. Back-Right B. Back
C. Front-Left D. **Left**



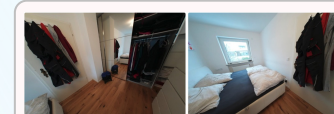
You are positioned at box and face clock. In which direction is bottle relative to you?
A. Back-Right B. **Left**
C. Back D. Front-Right



When you were taking the photo in Image B, where is the chair in Image A relative to you?
A. **Left-Front** B. Left-Back
C. Right-Back D. Right-Front

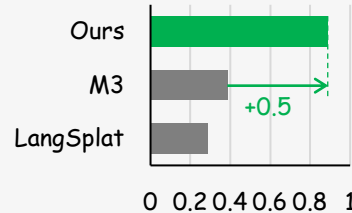


Given view A and view B, Which single rotation direction is the most prominent?
A. tile up B. **turn right**
C. turn left D. tilt down

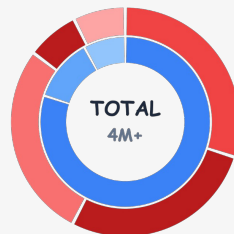
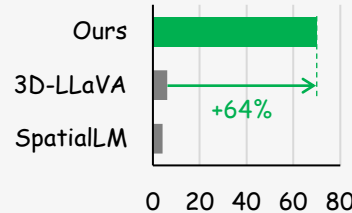


The direction of pillow relative to painting is south. What's direction of mirror relative to painting?
A. south-west B. north-east
C. **south** D. east

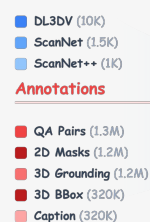
Depth F1 Score



3D Det. AP50



Datasets



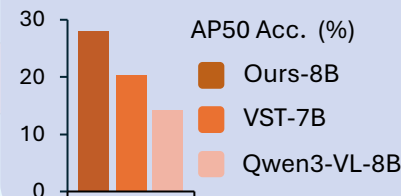
HOLI-SPATIAL-4M

VLM

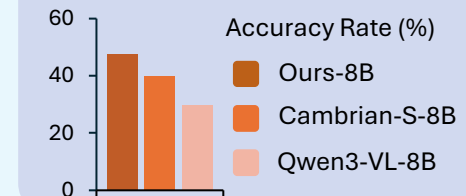
3D Grounding

Spatial Reasoning

Scannet++ 3D Grounding



MindeCube Spatial Reasoning



✓ High-Quality 3D Recon.

✓ Accurate Semantic Anno.

✓ Scalable Anno. Data